

Ease of setup: Amazon Managed Blockchain simplifies the process of creating a blockchain network, allowing users to set up a network with just a few clicks through the AWS management console.

Network management: Users can create and manage multiple blockchain networks, each with its own set of members and policies.

Security: The service integrates with AWS identity and access management (IAM) to control access to resources securely. It also allows for encryption of data in transit and at rest.

Scalability: Amazon Managed Blockchain can automatically scale to accommodate changes in network activity and transaction volume, ensuring high availability.

Monitoring and analytics: The service provides real-time monitoring and insights into the performance of the blockchain network, helping users identify and address issues quickly.

Smart contracts: It supports the deployment of smart contracts (Chaincode for Hyperledger Fabric and Solidity for Ethereum) to automate business processes.

Integration: Amazon Managed Blockchain can be integrated with other AWS services like Amazon S3, AWS Lambda, and Amazon CloudWatch for building comprehensive blockchain applications.

Amazon Quantum Ledger Database (Amazon QLDB)

While not a traditional blockchain, Amazon QLDB is a fully managed ledger database service designed to provide a transparent and immutable journal for tracking changes to data over time. Key features of Amazon QLDB include:

Immutable ledger: It maintains an immutable and tamper-evident record of all changes made to data, making it suitable for applications that require a transparent and auditable data history.

Data verification: The service provides cryptographic verification of data integrity, ensuring that data remains unchanged and can be trusted.

SQL-like queries: Users can query the ledger using a familiar SQL-like language, simplifying data retrieval and analysis.

Managed service: QLDB is fully managed by AWS, reducing operational overhead, and making it easy to scale as needed.

Integration: It can be integrated with other AWS services and applications to create comprehensive data-driven solutions.

AWS blockchain templates

AWS provides templates for popular blockchain frameworks, such as Ethereum and Hyperledger Fabric, to help users quickly deploy and configure blockchain networks on AWS infrastructure. These templates simplify the setup process.

Amazon Managed Blockchain for Ethereum

This service is specifically designed for Ethereum-based blockchain networks. It offers Ethereum-compatible APIs, smart contract support, and simplified management of Ethereum nodes.

Amazon Managed Blockchain for Hyperledger Fabric

This service is tailored for Hyperledger Fabric-based blockchain networks. It provides tools for creating, configuring, and managing Hyperledger Fabric nodes and networks.

Amazon Aurora with Amazon QLDB integration

AWS offers integration between Amazon Aurora (a relational database service) and Amazon QLDB, allowing businesses to create hybrid applications that combine the benefits of traditional databases with the immutability of ledger technology.

Ethereum on AWS

Ethereum's versatility has led to its widespread adoption for various use cases, including decentralised finance (DeFi), non-fungible tokens (NFTs), supply chain management, identity verification, and more. However, as with any technology, it's important to understand its capabilities and limitations before building or investing in projects on the Ethereum blockchain.

Key features of the Ethereum blockchain

Ethereum is a decentralised, open source blockchain platform that enables the creation and execution of smart contracts and decentralised applications (DApps). It was proposed in late 2013 by Vitalik Buterin and development began in early 2014, with the mainnet (live blockchain network) launched on July 30, 2015. The key features of the Ethereum blockchain include:

Smart contracts: Ethereum introduced the concept of smart contracts, which are self-executing contracts with the terms of the agreement directly written into code. These contracts automatically execute when specific conditions are met, without the need for intermediaries.

Decentralised applications (DApps): Ethereum allows developers to build decentralised applications on its platform. These DApps can serve a wide range of purposes, from financial services to gaming and more. They are typically open source and operate on the blockchain, providing transparency and security.

Ether (ETH): Ether is the native cryptocurrency of the Ethereum network. It is used to pay for transaction fees and computational services on the platform. It's also often seen as a digital asset and investment.

Decentralised autonomous organisations (DAOs): Ethereum supports the creation of DAOs, which are organisations run by code and the consensus of their members. They can